

Formulario

$$\nabla\phi = \sum_{i=1}^3 \mathbf{e}_i \frac{1}{h_i} \frac{\partial\phi}{\partial u_i}$$

$$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \left(\sum_{i=1}^3 \frac{\partial}{\partial u_i} (h_j h_k F_i) \right)$$

$$\nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \partial/\partial u_1 & \partial/\partial u_2 & \partial/\partial u_3 \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}$$

$$\nabla^2 \phi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_i} \left(\frac{h_j h_k}{h_i} \frac{\partial \phi}{\partial u_i} \right) \right]$$

$$h_x = 1, \quad h_y = 1, \quad h_z = 1$$

$$h_\rho = 1, \quad h_\varphi = \rho, \quad h_z = 1$$

$$h_r = 1, \quad h_\theta = r, \quad h_\varphi = r \sin \theta$$

$$\nabla(uv) = u\nabla v + v\nabla u, \quad \nabla \cdot (u\mathbf{A}) = u\nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla u$$

$$\nabla \times (u\mathbf{A}) = u\nabla \times \mathbf{A} - \mathbf{A} \times \nabla u, \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} + \mathbf{A} (\nabla \cdot \mathbf{B}) - (\mathbf{A} \cdot \nabla) \mathbf{B} - \mathbf{B} (\nabla \cdot \mathbf{A})$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

$$(\nabla \times \mathbf{A}) \times \mathbf{A} = (\mathbf{A} \cdot \nabla) \mathbf{A} - \nabla \left(\frac{A^2}{2} \right)$$

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\mathbf{r}_1 - \mathbf{r}_2|^3} (\mathbf{r}_1 - \mathbf{r}_2)$$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} q \frac{\mathbf{r} - \mathbf{r}_0}{|\mathbf{r} - \mathbf{r}_0|^3}, \quad \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int d^3 r' \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}$$

$$(\mathbf{E}_2 - \mathbf{E}_1) \cdot \hat{\mathbf{n}} = \frac{\sigma}{\epsilon_0}, \quad \hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0$$

$$U = \frac{1}{2} \sum_i q_i \Phi(\mathbf{r}_i), \quad U = \frac{1}{2} \int \rho(\mathbf{r}) \Phi(\mathbf{r}) d^3 r$$

$$\mathbf{x} = \phi \nabla \varphi$$

$$\int_V (\phi \nabla^2 \varphi + \nabla \phi \cdot \nabla \varphi) dV = \oint_S \phi \frac{\partial \varphi}{\partial n} dS$$

$$\int_V (\phi \nabla^2 \varphi + \varphi \nabla^2 \phi) dV = \oint_S \left(\phi \frac{\partial \varphi}{\partial n} - \varphi \frac{\partial \phi}{\partial n} \right) dS$$

$$\Phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V G \rho d^3 r' + \frac{1}{4\pi} \oint_S \left(G \frac{\partial \Phi}{\partial n'} - \Phi \frac{\partial G}{\partial n'} \right) dS'$$

$$[z - \mathcal{L}(\mathbf{r})] G(\mathbf{r}, \mathbf{r}'; z) = -4\pi \delta(\mathbf{r} - \mathbf{r}')$$

$$\nabla'^2 G(\mathbf{r}, \mathbf{r}') = -4\pi \delta(\mathbf{r} - \mathbf{r}'), \quad z = 0, \quad \mathcal{L} = \nabla'^2$$

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{|\mathbf{r} - \mathbf{r}'|} + F(\mathbf{r}, \mathbf{r}'), \quad \nabla'^2 F(\mathbf{r}, \mathbf{r}') = 0$$

$$G_D(\mathbf{r}, \mathbf{r}') = 0, \quad \mathbf{r}' \in S$$

$$\Rightarrow \Phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V G_D \rho d^3 r' - \frac{1}{4\pi} \oint_S \Phi \frac{\partial G_D}{\partial n'} dS'$$

$$\Phi(x, y, z) = X(x)Y(y)Z(z)$$

$$\Rightarrow \Phi(x, y, z) = e^{\pm i\alpha x} e^{\pm i\beta y} e^{\pm i\sqrt{\alpha^2 + \beta^2} z}$$

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \sum_{\ell=0}^{\infty} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} P_{\ell}(\cos \gamma)$$

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = 4\pi \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{1}{2\ell+1} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} Y_{\ell m}^*(\theta', \phi') Y_{\ell m}(\theta, \phi)$$

$$r_{<} = \min(|\mathbf{x}|, |\mathbf{x}'|), \quad r_{>} = \max(|\mathbf{x}|, |\mathbf{x}'|)$$

$$\oint Y_{\ell m}^*(\Omega) Y_{\ell m}(\Omega) d\Omega = \delta_{\ell\ell'} \delta_{mm'}$$

$$\delta^{(3)}(\mathbf{r} - \mathbf{r}') = \frac{\delta(r - r')}{r^2} \sum_{\ell, m} Y_{\ell}^m(\hat{\mathbf{r}}) Y_{\ell}^{m*}(\hat{\mathbf{r}}')$$

$$G(r, r') = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} g_{\ell}(r, r') P_{\ell}(\cos \gamma)$$

$$\left. \frac{dg_{\ell}}{dr} \right|_{r'+} - \left. \frac{dg_{\ell}}{dr} \right|_{r'-} = -\frac{4\pi}{r'^2}$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int d^3 r' \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}, \quad \mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint_C d\mathbf{l}' \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}$$

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0, \quad \hat{\mathbf{n}} \times (\mathbf{B}_1 - \mathbf{B}_2) = \mu_0 \mathbf{K}$$

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int d^3 r' \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

$$\mathbf{J}_M = \nabla \times \mathbf{M}, \quad \mathbf{K}_M = \mathbf{M} \times \hat{\mathbf{n}}, \quad \rho_M = -\nabla \cdot \mathbf{M}, \quad \sigma_M = \mathbf{M} \cdot \hat{\mathbf{n}}$$

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int_V \frac{\nabla' \times \mathbf{M}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3 x' + \frac{\mu_0}{4\pi} \oint_S \frac{\mathbf{M}(\mathbf{x}') \times \hat{\mathbf{n}}'}{|\mathbf{x} - \mathbf{x}'|} da'$$

$$\mathbf{J}_f = 0 \quad \Leftrightarrow \quad \mathbf{H} = -\nabla \Phi_M, \quad \nabla^2 \Phi_M = -\rho_M$$

$$\Phi_M = -\frac{1}{4\pi} \int_V \frac{\nabla' \cdot \mathbf{M}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3 x' + \frac{1}{4\pi} \oint_S \frac{\hat{\mathbf{n}}' \cdot \mathbf{M}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} da'$$

$$\mathbf{A} \approx \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3}, \quad \mathbf{m} = \frac{1}{2} \int d^3 r' \mathbf{r}' \times \mathbf{J}(\mathbf{r}')$$

$$\mathbf{N} = \mu \times \mathbf{B}, \quad \mu = I\mathbf{S}, \quad U = -\mathbf{m} \cdot \mathbf{B}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \int_V \rho d^3 r, \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0, \quad \oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{a} + \frac{1}{c^2} \int_S \mathbf{E} \cdot d\mathbf{a}$$

$$\mu_0 \epsilon_0 = \frac{1}{c^2}, \quad \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad \mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$$

$$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{H} = \frac{1}{\mu} \mathbf{B}$$

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \nabla \times \mathbf{E} = \frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla \Phi - \frac{\partial \mathbf{A}}{\partial t}$$

$$\nabla^2 \Phi + \frac{\partial \mathbf{A}}{\partial t} = -\frac{\rho}{\epsilon_0}$$

$$\nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \Phi}{\partial t} \right) = -\mu_0 \mathbf{J}$$

$$\mathbf{A}' = \mathbf{A} + \nabla \Lambda, \quad \Phi' = \Phi - \frac{\partial \Lambda}{\partial t}$$

$$\nabla \cdot \mathbf{A}' + \frac{1}{c^2} \frac{\partial \Phi'}{\partial t} = 0$$

$$\nabla^2 \Lambda - \frac{1}{c^2} \frac{\partial^2 \Lambda}{\partial t^2} = - \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \Phi}{\partial t} \right)$$

$$\Rightarrow \nabla^2 \Phi - \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2} = -\frac{\rho}{\epsilon_0}, \quad \nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}$$

$$\nabla \cdot \mathbf{A}' = 0$$

$$\nabla^2 \Lambda = -\nabla \cdot \mathbf{A}$$

$$\mathbf{J} = \mathbf{J}_\ell + \mathbf{J}_t, \quad \nabla \times \mathbf{J}_\ell = 0, \quad \nabla \cdot \mathbf{J}_t = 0$$

$$\Rightarrow \nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}_t, \quad \nabla^2 \Phi = -\frac{\rho}{\epsilon_0}$$

$$\mathbf{J}_\ell = \epsilon_0 \nabla \frac{\partial \Phi}{\partial t}$$

$$\hat{\mathbf{R}} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}, \quad R = |\mathbf{r} - \mathbf{r}'|, \quad t' = t_{\text{ret}} = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}$$

$$\Phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{1}{R} \rho(\mathbf{r}', t_{\text{ret}})$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int d^3r' \frac{1}{R} \mathbf{J}(\mathbf{r}', t_{\text{ret}})$$

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{1}{R^2} \left[\left(\rho + \frac{R}{c} \frac{\partial}{\partial t'} \rho \right) \hat{\mathbf{R}} - \frac{1}{c} \frac{\partial}{\partial t'} \mathbf{J} \right]$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int d^3r' \frac{1}{R^2} \left[\mathbf{J} + \frac{R}{c} \frac{\partial}{\partial t'} \mathbf{J} \right] \times \hat{\mathbf{R}}$$

$$g_\omega^{(\pm)}(R) = \frac{e^{\pm i k R}}{R}, \quad G^\pm = \frac{1}{R} \delta \left(\tau \mp \frac{R}{c} \right)$$

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \psi(\mathbf{r}, t) = -4\pi f(\mathbf{r}, t)$$

$$\Rightarrow \psi^{(\pm)}(\mathbf{r}, t) = \psi_{\text{hom}}(\mathbf{r}, t) + \int d^3r' \int dt' G^{(\pm)}(\mathbf{r}, t; \mathbf{r}', t') f(\mathbf{r}', t')$$

$$\Rightarrow \psi^+(\mathbf{r}, t) = \int \frac{[f(\mathbf{r}', t)]_{\text{ret}}}{|\mathbf{r} - \mathbf{r}'|} d^3r' \quad \Leftrightarrow \quad \psi_{\text{in}}(\mathbf{r}, t) = 0$$

$$\Phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\kappa R} \right]_{\text{ret}}, \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 q}{4\pi} \left[\frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}}$$

$$\beta = \mathbf{v}/c, \quad \kappa = 1 - \hat{\mathbf{R}} \cdot \beta$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{v}_{\text{ret}}}{c^2} \Phi(\mathbf{r}, t)$$

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \left[\frac{(1 - \beta^2)(\hat{\mathbf{R}} - \beta)}{\kappa^3 R^2} + \frac{\hat{\mathbf{R}} \times \left[(\hat{\mathbf{R}} - \beta) \times \dot{\beta} \right]}{c\kappa^3 R} \right]_{\text{ret}}$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \left[\hat{\mathbf{R}} \times \mathbf{E} \right]_{\text{ret}}$$

$$u = \frac{\epsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2, \quad \mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$$

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$$

$$u = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H}), \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}$$

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}, \quad \mathbf{g} \equiv \epsilon_0 \mathbf{E} \times \mathbf{B} = \frac{1}{c^2} \mathbf{S}$$

$$T_{\alpha\beta} = \epsilon_0 \left(E_\alpha E_\beta - \frac{1}{2} E^2 \delta_{\alpha\beta} \right) + \frac{1}{\mu_0} \left(B_\alpha B_\beta - \frac{1}{2} B^2 \delta_{\alpha\beta} \right)$$

$$\frac{\partial \mathbf{g}}{\partial t} + \mathbf{f} = \nabla \cdot \mathbf{T}, \quad T_{\alpha\beta} = T_{\beta\alpha}$$

$$\mathbf{L}_{\text{field}} = \int \mathbf{r} \times \mathbf{g} d^3r$$

$$\mathbf{r} \xrightarrow{\mathcal{P}} -\mathbf{r}, \quad \mathbf{E} \xrightarrow{\mathcal{P}} -\mathbf{E}, \quad \mathbf{B} \xrightarrow{\mathcal{P}} \mathbf{B}, \quad \Phi \xrightarrow{\mathcal{P}} \Phi, \quad \mathbf{A} \xrightarrow{\mathcal{P}} -\mathbf{A}$$

$$\mathbf{r} \xrightarrow{\mathcal{R}} R\mathbf{r}, \quad \mathbf{E} \xrightarrow{\mathcal{R}} R\mathbf{E}, \quad \mathbf{B} \xrightarrow{\mathcal{R}} R\mathbf{B}$$

$$t \xrightarrow{\mathcal{T}} -t, \quad \mathbf{r} \xrightarrow{\mathcal{T}} \mathbf{r}, \quad \mathbf{J} \xrightarrow{\mathcal{T}} -\mathbf{J}, \quad \rho \xrightarrow{\mathcal{T}} \rho, \quad \mathbf{E} \xrightarrow{\mathcal{T}} \mathbf{E}, \quad \mathbf{B} \xrightarrow{\mathcal{T}} -\mathbf{B}$$

$$\int f(x) \delta(x-a) dx = f(a), \quad \int f(x) \delta'(x-a) dx = -f'(a)$$

$$\partial_i \partial_j \left(\frac{1}{r} \right) = \frac{3x_i x_j - r^2 \delta_{ij}}{r^5} - \frac{4\pi}{3} \delta_{ij} \delta(\mathbf{r})$$

$$\mathbf{p}(\mathbf{r}) = \int_V \rho(\mathbf{r}') (\mathbf{r} - \mathbf{r}') d^3r', \quad \mathbf{p}(\mathbf{r}) = \sum_{i=1}^N q_i (\mathbf{r}_i - \mathbf{r})$$

$$f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

$$f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + \dots$$

$$P_0(x) = 1,$$

$$Y_0^0(\theta, \varphi) = \sqrt{1/4\pi}$$

$$P_1(x) = x,$$

$$Y_1^{-1}(\theta, \varphi) = \sqrt{3/8\pi} e^{-i\varphi} \sin \theta$$

$$P_2(x) = (3x^2 - 1)/2,$$

$$Y_1^0(\theta, \varphi) = \sqrt{3/4\pi} \cos \theta$$

$$P_3(x) = (5x^3 - 3x)/2,$$

$$Y_1^1(\theta, \varphi) = -\sqrt{3/8\pi} e^{i\varphi} \sin \theta$$